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Theoretical Foundations

1.1 | Principal Component Analysis

Principal Component Analysis (PCA) is a technique that transforms a large set of variables into a smaller one that still contains most of the information in the original set by identifying the directions along which the data varies the most. As we will see, this is achieved by projecting the data onto the eigenvectors of the covariance matrix of the data, where these eigenvectors, or principal components, correspond to the directions of maximum variance, and the associated eigenvalues represent the magnitude of variance in these directions. This can be extremely helpful when dealing with high-dimensional data where the goal is dimensionality reduction, data compression, feature extraction or visualization of these high-dimensional data.

The first description of PCA can be traced back to Pearson [19] in 1901 and Hotelling [16] in 1936, but it was not until 1956 when Lorenz [18] popularized PCA in the field of meteorology under the term *empirical orthogoanal function analysis*¹ Subsequently, PCA has been widely used in meteorology and climate science to identify the dominant modes of variability in the atmosphere and oceans with entire chapters of popular textbooks being dedicated to this technique [e.g. 12, 20, 23].

In the following we will quickly derive PCA from the maximum variance formulation², [e.g. 2, pp. 561-563], in which the data is orthogonally projected onto a lower dimensional linear space, known as the principal subspace, such that the variance of the projected data is maximized. Then we will discuss the main shortcomings of PCA in particular in the context of climate science and some of the most common ways to overcome these.

1.1.1 Derivation based on Maximum Variance Formulation

Consider a data set represented as a two dimensional $N \times D$ matrix $\mathbf{X}$ of N observations with D features such that the element x_{ni} represents the n^{th} observations of the i^{th} feature.

¹Indeed, until today EOF analysis seems to be the preferred term used in meteorology and climate science. We will stick to the more commonly used term PCA though.

²This is what Hotelling [16] used to derive PCA. An alternative derivation is the minimum error formulation by Pearson [19], in which we seek a linear projection that minimizes the average projection cost, defined as the mean squared distance between the data points.

of observations $\{\mathbf{x}_n\}$ where $n = 1, \dots, N$, and $\mathbf{x}_n$ is a Euclidean variable with dimensionality D . Our goal is to project the data onto a space having dimensionality $M < D$ while maximizing the variance of the projected data. For the moment, we shall assume that the value of M is given. Later in this chapter, we shall consider techniques to determine an appropriate value of M from the data.

To begin with, consider the projection onto a one-dimensional space ($M = 1$). We can define the direction of this space using a D -dimensional vector $\mathbf{v}_1$, which for convenience we shall choose to be a unit vector so that $\mathbf{v}_1^T \mathbf{v}_1 = 1$. Note that we are only interested in the direction of $\mathbf{v}_1$, not its magnitude. Each data point $\mathbf{x}_n$ is then projected onto a scalar value $\mathbf{v}_1^T \mathbf{x}_n$, and the variance of the projected data is given by

$$\bar{\mathbf{x}} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{x}_n \quad (1.1)$$

and the variance of the projected data is given by

$$\frac{1}{N-1} \sum_{n=1}^N \{\mathbf{v}_1^T \mathbf{x}_n - \mathbf{v}_1^T \bar{\mathbf{x}}\}^2 = \mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 \quad (1.2)$$

where $\mathbf{S}$ is the covariance matrix defined by

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T.$$

We now maximize the projected variance $\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. Since the maximization problem does not have a bounded (unique) solution, we need constrained maximization using our normalization condition from above $\mathbf{v}_1^T \mathbf{v}_1 = 1$. To enforce this constraint, we introduce a Lagrange multiplier that we shall denote by λ_1 , and then make an unconstrained maximization of

$$\mathbf{v}_1^T \mathbf{S} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^T \mathbf{v}_1).$$

By setting the derivative with respect to $\mathbf{v}_1$ equal to zero, we see that this quantity will have a stationary point when

$$\mathbf{S} \mathbf{v}_1 = \lambda_1 \mathbf{v}_1.$$

which says that $\mathbf{v}_1$ must be an eigenvector of $\mathbf{S}$ and λ_1 the corresponding eigenvalue. If we left-multiply by $\mathbf{v}_1^\top$ and make use of the normalization condition, we see that the variance is given by

$$\mathbf{v}_1^\top \mathbf{S} \mathbf{v}_1 = \lambda_1$$

and so the variance will be a maximum when we set $\mathbf{v}_1$ equal to the eigenvector having the largest eigenvalue λ_1 . The eigenvector is known as the first *principal component*³ of the data set.

We can define additional principal components in an incremental fashion by choosing each new direction to be that which maximizes the projected variance amongst all possible directions orthogonal to those already considered. If we consider the general case of an M -dimensional projection space, the optimal linear projection for which the variance of the projected data is maximized is now defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the data covariance matrix $\mathbf{S}$ corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This is easily shown using proof by induction (Appendix A), yielding the concise expression

$$\mathbf{S}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}, \quad (1.3)$$

where by construction the eigenvector matrix $\mathbf{V}$ is orthogonal, i.e. $\mathbf{V}^\top \mathbf{V} = \mathbf{I}_M$.

Thus we see how the eigenvectors of $\mathbf{S}$ arise naturally when we try to find a new coordinate system "along" regularities in the data. For what to do when the regularities in the data are not aligned in the orthogonal directions, see Section 1.1.2.

From the spectral theorem for normal matrices [e.g. 15, p. 133] it follows that the symmetric covariance matrix $\mathbf{S}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding eigenvalues λ_i :

$$\mathbf{S} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^\top. \quad (1.4)$$

Therefore, each eigenvalue λ_i represents the amount of variance described by each component which is typically expressed in form of a fraction γ_i for the corresponding i^{th} component:

³Naming convention depends a lot on the scientific domain. For instance, the pair of eigenvectors (here $\mathbf{V}$) and corresponding data projections (here $\mathbf{U}$) are called the principal components and principal component scores in machine learning [e.g. 2], vector of coefficients/loadings and principal components in statistics [e.g. 17] or empirical orthogonal functions and principal components/expansion coefficients in climate science and meteorology [e.g. 23]. Here I try to be consistent and follow the convention used in machine learning.

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.5)$$

Projecting the data $\mathbf{X}$ onto the M dimensional subspace thus provides the data representation in the projection space, called the raw principal component scores

$$\tilde{\mathbf{U}} = \mathbf{X}\mathbf{V}, \quad (1.6)$$

which gives also the possibility to reconstruct the original data from (a subset of) the projected data:

$$\mathbf{X} = \tilde{\mathbf{U}}\mathbf{V}^T. \quad (1.7)$$

As we will see later, computing the eigenvector decomposition is not the most efficient way computationally seen. A more efficient way to compute the eigenvectors involves the singular value composition (SVD) [e.g. 15, p. 150] of the data matrix

$$\mathbf{X} = \mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^T, \quad (1.8)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are respectively $N \times M$ and $M \times D$ unitary matrices, i.e. $\mathbf{U}'^T\mathbf{U}' = \mathbf{V}'^T\mathbf{V}' = \mathbf{I}_M$ where $M \leq \min(N, D)$ is the rank of $\mathbf{X}$ and $\mathbf{I}_M$ is the identity matrix of order M . The matrix $\mathbf{\Sigma}$ is a diagonal matrix of the singular values σ_i of $\mathbf{X}$, i.e. $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_M \geq 0$.

The covariance matrix $\mathbf{S}$ can be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}^T \mathbf{X} = \frac{1}{N-1} \mathbf{V}' \mathbf{\Sigma}^T \mathbf{U}'^T \mathbf{U}' \mathbf{\Sigma} \mathbf{V}'^T = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T, \quad (1.9)$$

where, by comparison with (1.3), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1} \mathbf{\Sigma}^T \mathbf{\Sigma}$. This last equation directly links the eigenvalues of the covariance matrix to the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.10)$$

Comparing (1.7) with (1.8) shows that the raw principal component scores $\tilde{\mathbf{U}}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \tilde{\mathbf{U}}\boldsymbol{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \tilde{\mathbf{U}}\boldsymbol{\Lambda}^{-1/2}, \quad (1.11) \quad 1$$

so that $\mathbf{U}^T \mathbf{U} = \mathbf{I}_M$ and the principal component scores $\mathbf{U}$ are unitary, thus uncorrelated. 2

Computational considerations Computationally, solving the eigendecomposition of the covariance matrix scales with $O(D^3)$ making it infeasible for modern data sizes in e.g. climate science where typical data sets⁴ are of the order $D = O(10^6)$, but can be even much larger in practice. 3
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In practice, alternative, more efficient algorithms involving the singular value decomposition (SVD) or the power methods [9] have been used instead from the beginning of the 21st century onward. More recently, randomized low-rank approximation techniques [8, 10] have allowed to considerably reduce the computational complexity to $O(ND \log(M) + (N + D)M^2)$ offering feasible alternatives for large, out-of-memory datasets (Table 1.1). 6
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Shortcomings Despite its popularity and wide applicability, PCA has several limitations and potential drawbacks that should be considered when using this technique: 11
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1. **Linearity assumption:** PCA assumes that the data's underlying structure is linear, which may not always hold true, especially in complex systems like climate data where nonlinear relationships can be significant. 13
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2. **Interpretability of principal components:** In addition to the linearity assumption, the dominant modes of variability are represented in an orthogonal subspace spanned by the principal components. This orthogonality is a mathematical construction that may not always correspond to the physical reality of the data, potentially leading to misinterpretations of the results (see Section 1.1.2). 16
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3. **Variance focus:** Capturing the directions of maximum variance may not always correspond to the most important features for a given application, potentially overlooking critical but less variable information. 21
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4. **Static nature:** As a static method PCA does not account for temporal dynamics or changes over time, which can be particularly limiting in climate science where time evolution is a key aspect. 24
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5. **Sensitivity to outliers:** PCA can be sensitive to outliers, as extreme values can disproportionately influence the principal components, leading to misleading results. 26
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⁴For instance, a single climate variable of ERA5, the go-to standard reference for state-of-the-art reanalysis climate data is distributed on a regular 0.25° grid of Earth surface which translates to $D = 1\,036\,800$ grid cells.

Table 1.1 Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [2] and Halko *et al.* [10].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

6. **Sensitivity to scaling:** The results are sensitive to the scaling of the input data, meaning that variables with larger ranges can dominate the principal components unless the data is appropriately standardized.

7. **Loss of information:** Not a shortcoming by itself because it's sometime the goal of PCA, but it should be taken into account that by reducing dimensionality, PCA inherently discards some information, which might lead to the loss of important details that could be crucial for accurate analysis and interpretation in specific applications.

Over the last decades a plethora of modifications and extensions have been proposed to overcome these limitations. We only mention a few here that are particularly relevant in the context of climate science. These include various forms of kernel PCA [4, 22, 25] to adress the linearity assumption, rotated [21] and sparse [11] PCA for interpretability, robust PCA [5] to address the sensitivity to outliers, and extended [3, 24] and complex [1, 14] PCA to account for temporal dynamics. Although not strictly a form of PCA, eigenmethods like optimal persistence analysis (OPA) [6, 7], which allows to filter for spatial patterns linked to increased persistence, and Principal Oscillation Pattern (POP) analysis [13] that provides a mean to analyze dynamical systems, are an example of related appraoches with the goal to gain insights about the dominant pattern of a physical field in climate science.

In the following sections, we will discuss some of the shortcomings and proposed solutions in more detail: in particular we will discuss how an additional rotation of the PCA solution can help address the challenge of physical interpretability (Section 1.1.2) and how complexifying the data provides a means to better understand time evolution patterns (Section 1.1.3).

1.1.2 Rotated PCA

The physical interpretation of the principal componetns is limited by two fundamental properties of PCA.

First, while it is often possible to clearly associate the first components with a known physical process, this is much more difficult with the second (and higher-order) components because it is

constrained to be orthogonal to the first component. However, real-world processes do not need to have orthogonal patterns or uncorrelated scores.

Secondly, while beneficial in many cases, the ability for efficient data compression inherent in the dimensionality reduction of PCA typically leads to the principal components being dense, i.e. all elements of the eigenvectors are non-zero. This can make the interpretation of the principal components difficult.

It is mainly due to these two properties that lead to a variety of undesired effects in the context of climate science. For instance, Richman [21] describes four effects, namely domain shape dependence, subdomain instability, sampling errors and non-intuitive patterns. The first two are a direct consequence of the orthogonality constraint, which for rectangular domains – like a regular longitude-latitude grid – leads to a set of patterns known as the Buell patterns [21] consisting of a positive pattern for the first component, subsequently followed by increasing polar i.e. bipole, tripole, quadrupole etc. patterns for the subsequent components (Figure 1.1), which upon shifting the domain can lead to a completely different set of patterns.

The third effect is related to fact that finite sample sizes will always lead to a sampling error [north_sampling_1982] which can be severe when eigenvalues are closely spaced.

The fourth effect is a logical conclusion from the first three, but which is enhanced by the dense nature of the principal components which is often not representative of the underlying physical processes, which are often more regionalized.

To address these limitations, rotated PCA techniques [e.g. 21] have been developed with the goal to produce "simpler" patterns in the hope that these would automatically better align with the underlying physical processes.

This simplicity is vaguely defined in a heuristic manner, but typically refers to patterns that are more localized, less dense, and more physically interpretable.

Thurstone states that all of the common variance of a variable should be accounted for by fewer than r common factors.

The underlying idea is to find a rotation matrix $\mathbf{R}$ that rotates the principal components $\mathbf{V}$ to a new set of components $\mathbf{V}_R$ that are more physically interpretable.

nicrie: drawbacks of rotation, e.g. in Storch, alternatives based on lasso; problem though with computation complexity

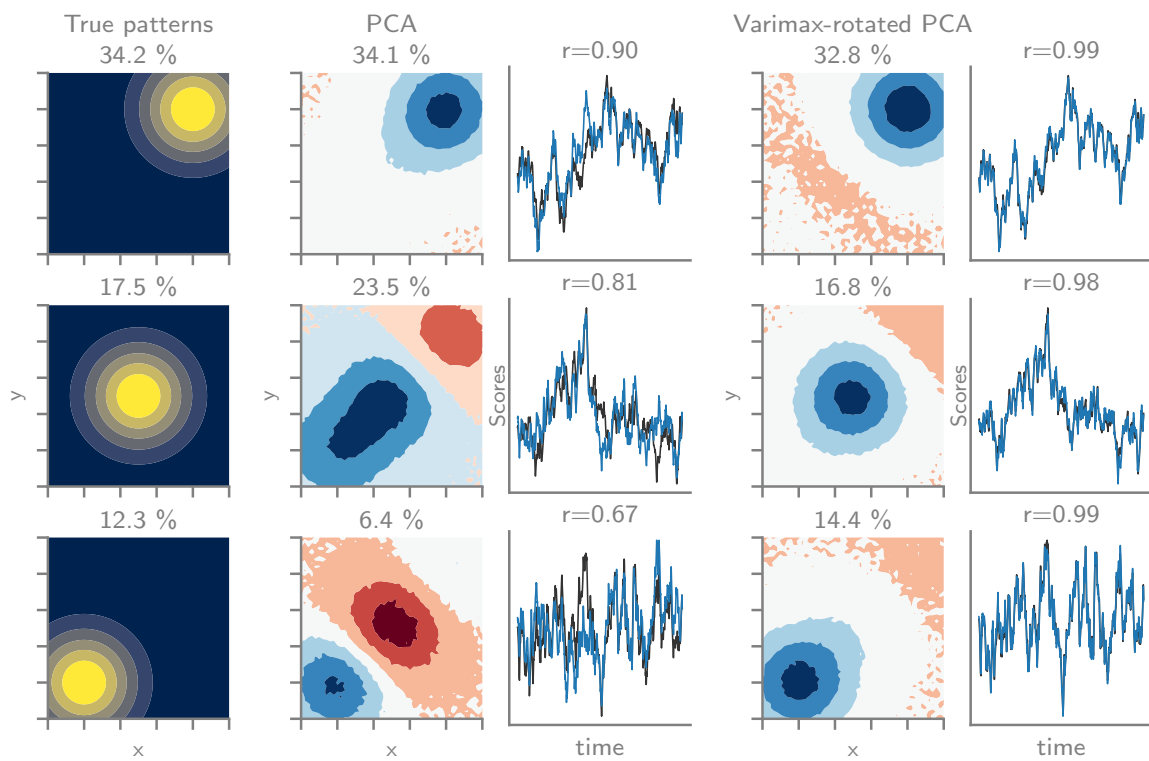


Fig. 1.1 Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

1.1.3 Hilbert PCA

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1.1.4 Probabilistic PCA

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nicrie: PCA arises naturally as the maximum likelihood solution a particular form of linear-Gaussian latent variable model. assumption 1: the simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables assumption 2: make use of linear, Gaussian dependence of the observed variables on the slate of the latent variables. assumption 1+ 2 give rise to Probabilistic PCA

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Finally, it is worth noting that there exists a closely related linear dimensionality reduction technique called canonical correlation analysis, or CCA (Hotelling, 1936; Bach and Jordan, 2002). Whereas PCA works with a single random variable, CCA considers two (or more) variables and tries to find a corresponding pair of linear subspaces that have high cross-correlation, so that each component within one of the subspaces is correlated with a single component from the other subspace. Its solution can be expressed in terms of a generalized eigenvector problem.

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1.2 | Maximum Covariance Analysis

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1.2.1 Rotated Maximum Covariance Analysis

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1.2.2 Complex Maximum Covariance Analysis

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1.3 | Canonical Correlation Analysis

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1.3.1 Complex Canonical Correlation Analysis

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1.4 | Neural Networks

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1.4.1 Introduction to Neural Networks

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Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

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1.4.2 Recurrent Neural Networks

Introduction to RNNs⁵: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM⁶: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.4.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

Layerwise Relevance Propagation (LRP)

Concept of LRP⁷: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP⁸: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

⁵Graves 2011 Supervised Sequence Labelling with Recurrent Neural Networks

⁶Hochreiter & Schmidhuber 1997

⁷Bach et al. 2015

⁸Lundberg 2017

1.5 | Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

1.5.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

1.5.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

1.5.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1 **1.5.4 Model Implementation and Inference**

- 2 Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial
3 distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and
4 Prediction: Explain how to perform inference and make predictions using the fitted model.

5 **1.6 | Comparative Perspective**

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Proofs

Deriving PCA for the general M -dimensional projection space

Here we generalize the solution for the optimal linear projection for the case of an M -dimensional projection space. In Chapter 1.1, we have already shown that for $M = 1$, the variance of the projected data is maximized when $\mathbf{u}_1$ is the eigenvector of the covariance matrix $\mathbf{S}$ corresponding to the largest eigenvalue λ_1 .

Using proof by induction we will show that the optimal linear projection for an $(M - 1)$ dimensional subspace is given by the $M - 1$ eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$ corresponding to the $M - 1$ largest eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{M-1}$.

First, we need to find $\mathbf{u}_M$ such that the variance of the projection onto the M -dimensional subspace is maximized. The total variance for the M -dimensional projection space can be expressed as the sum of variances in each direction

$$\sum_{i=1}^M \mathbf{u}_i^T \mathbf{S} \mathbf{u}_i$$

where $\mathbf{u}_M$ must be orthogonal to the previously chosen eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{M-1}$:

$$\mathbf{u}_i^T \mathbf{u}_j = \delta_{ij} \quad \text{for } i, j = 1, 2, \dots, M.$$

Using the same methods as for the $M = 1$ case, we introduce a Lagrange multiplier λ_M and maximize

$$\mathbf{u}_M^T \mathbf{S} \mathbf{u}_M + \lambda_M (1 - \mathbf{u}_M^T \mathbf{u}_M).$$

Setting the derivative with respect to $\mathbf{u}_M$ to zero, we obtain

$$\mathbf{S} \mathbf{u}_M = \lambda_M \mathbf{u}_M$$

showing that $\mathbf{u}_M$ is an eigenvector of $\mathbf{S}$ with eigenvalue λ_M .

